A Spatial Contexts-Informed Self-Supervised Learning Approach for Pavement Distress Segmentation

Ruiqi Ren, Peixin Shi, and Jinwoo Kim

***Abstract*—Detection and repair of pavement distress in time are crucial to maximize functional performance and service life, while minimizing maintenance costs on extensive roadway networks. Manual distress detection is labor intensive and error prone. While deep learning techniques offer unparalleled capabilities for automated and accurate pixel-level pavement distress segmentation, their reliance on extensive manual annotations remains a bottleneck. To address this challenge, we propose an open-ended self-supervised framework enabling flexible integration of various pretext tasks for pavement distress segmentation without manual annotations. We introduce a spatial contexts-informed pretext task that automatically generates pseudo labels by leveraging the highly consistent semantic information inherent across continuous pavement images within localized areas. A multi-line parallel network architecture is then employed, where each line extracts a distinct deep representation aligned with the pseudo-label generation process. These representations are jointly optimized through a shared weight update scheme augmented by momentum encoders to capture long-range dependencies. A vision transformer processes the input images during inference, utilizing self-attention to highlight distressed regions based on the learned representations for precise segmentation. Extensive evaluations validate the performance of our framework, outperforming state-of-the-art self-supervised methods by 0.075 mIoU on average, while remarkably surpassing weakly supervised techniques requiring manual image-level annotations. These results are far more promising given that our self-supervised approach avoids human labeling costs, striking a trade-off between model effectiveness and annotation efficiency for large-scale deployments. It helps transportation agencies to realize timely, proactive infrastructure maintenance through scalable, accurate distress monitoring over extensive road networks.**

***Index Terms*—Pavement distress segmentation, self-supervised learning, spatial contexts, multi-line parallel networks.**

I. Introduction

DETECTION and repair of pavement distress in time are crucial to maximize functional performance and service life, while minimizing maintenance costs on extensive roadway networks. Manual visual distress inspection is labor-intensive, time-consuming, and error-prone. Digital and automated distress detection technologies, such as computer vision, thermal imaging technology and laser scanning, have become a promising solution [1], [2]. Among emerging technologies, computer vision–based methods have attracted significant attention in recent years [3], [4], as pavement distresses are typically expressed through visual cues that signal early stages of road deterioration. Currently, vehicles mounted with optical cameras are widely used in practice to regularly inspect pavement, locate and evaluate the distresses through computer vision technologies, and then provide information to support maintenance decisions [5], [6].

Supervised deep learning have become the core engine of computer vision-based pavement distress detection due to their high accuracy and strong generalization performance [7]. Although supervised learning shows considerable potential, its reliance on large volumes of training images and labor-intensive annotations limits widespread use in pavement distress detection [8], [9], [10]. This challenge is especially pronounced in segmentation tasks, where pixel-level annotations are required [11], [12]. Potential solutions to mitigate this challenge include data augmentation techniques [13] and transfer learning [14], [15]. While these approaches have demonstrated enhanced performance in pavement distress detection and segmentation, they cannot entirely bypass the requirement for manual annotations, given that each roadway has unique surrounding environments. Therefore, exploring methods to significantly lower or completely eliminate the requirement for manual annotations has become a worthwhile endeavor [16].

Training deep learning models without access to ground truth labels poses a fundamental challenge in defining effective learning objectives. Researchers have developed various approaches to reduce manual annotation requirements to varying degrees. Weakly supervised learning methods utilize coarse-grained labels (such as image-level annotations) to guide a learning process. They often learn a distribution of normal data and identify anomalies as instances that exhibit statistical deviations [17], [18], [19], [20], [21], [22]. These methods significantly lower the annotation burden compared to fully supervised approaches, but still require a substantial amount of labeled data.

Self-supervised learning eliminates the need for explicit annotations by leveraging data structure to generate pseudo-labels through pretext tasks. For example, models learn feature representations by reconstructing masked patches or aligning

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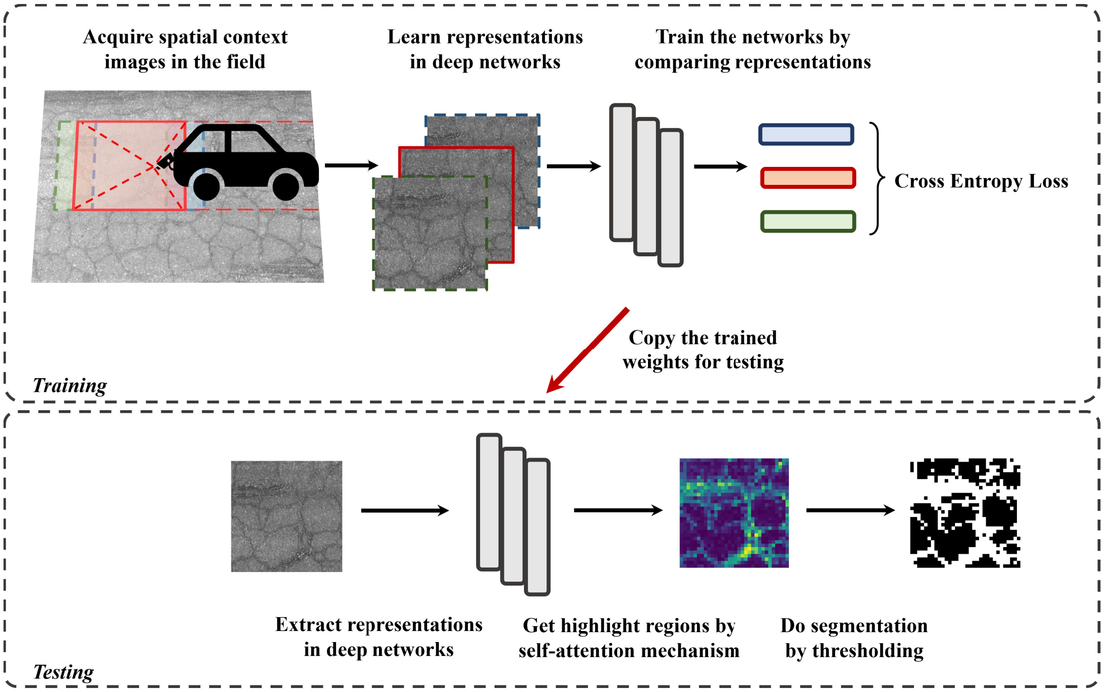


Fig. 1. Training schema of unsupervised siamese networks for pavement distress segmentation.

augmented views [23], [24], [25], [26]. However, these methods struggle to extract features of small pavement distress. Masked patches may obscure critical details, while similar pavement backgrounds make it difficult to optimize deep representation learning based on similarity. Self-supervised approaches also prioritize instance-level discrimination over dense predictions, limiting their effectiveness for precise localization or segmentation. Designing more suitable pretext tasks is essential for accurate pavement distress segmentation.

To achieve pixel-level pavement distress segmentation without manual annotations, we propose a spatial contexts-informed self-supervised learning approach for pixel-level pavement distress segmentation, as shown in Figure 1. A spatial contexts-informed pretext task leverages consistent semantic information across continuous pavement images to automatically generate pseudo-labels. A multi-line parallel network architecture is then trained on these pseudo-labeled inputs, updating parameters through shared weights and momentum encoders. Finally, pavement distress segmentation is achieved by the self-attention mechanism of the vision transformer (ViT).

II. Related Work

This section reviews relevant literature in pavement distress segmentation, encompassing three primary learning paradigms, supervised deep learning techniques that leverage fully annotated datasets, weakly supervised learning methods which exploit partial annotations to reduce manual labeling efforts, and self-supervised learning approaches that aim to extract meaningful representations from unlabeled data through carefully designed pretext tasks.

A. Supervised Deep Learning for Pavement Distress Segmentation

The advent of convolutional neural networks, notably U-Net [27] and fully convolutional networks (FCN) [28], has significantly influenced pavement distress detection, primarily through supervised learning approaches. Crack is the most commonly occurred and complex distress form, and has received extensive attention in academia. For instance, Zhang et al. enhance the U-Net by incorporating a generative adversarial network. This modification forces the network to segment images with fine crack details, thus improving segmentation precision [13]. Wang et al. introduce a rectangular convolution pyramid and edge enhancement network (RENet) for precise small crack segmentation [29]. Liu et al. propose CrackFormer, featuring a pyramid-shaped transformer structure and customized scaling attention for sharpened crack detection [11]. To mitigate the limitation of training data scarcity in supervised methods, Hou et al. achieve significant segmentation performance improvements by utilizing Wasserstein generative adversarial network-gradient penalty (WGAN-GP) for image enhancement before crack segmentation [30]. Zheng et al. manage to reduce the need for labeled images through active learning [15]. Despite these significant contributions, the persistent need for extensive manual annotation remains a major obstacle, impeding the widespread adoption of computer vision technologies in pavement distress detection.

B. Weakly Supervised Learning to Reduce Annotation Dependency

Efforts to mitigate the reliance on extensive manual annotations have led to the development of weakly supervised learning approaches. These methods often leverage image-level annotations, assuming the majority of data represents normal conditions. One approach trains encoder-decoder networks exclusively on normal instances, enabling accurate reconstruction of normal data. When presented with abnormal inputs, these networks struggle to reconstruct anomalous regions, leading to discernible deviations. By comparing original and reconstructed images, anomalies can be localized without pixel-level annotations. For instance, GANomaly employs an encoder-decoder architecture with a discriminator for anomaly detection in luggage [17], while f-AnoGAN explores W-GAN for medical imaging [18]. Wang et al. enhanced this structure for pavement distress detection by incorporating modules that comprehend fundamental texture information [31]. Yu et al. proposed FastFlow, utilizing 2D normalizing flows as a probability distribution estimator [32]. However, the direct application of these methods to pavement distress segmentation may be suboptimal due to the non-standardized nature of field-acquired pavement images and the prevalence of distressed sections in typical data collection practices.

Another weakly supervised learning scheme tries to establish a mapping between different image categories. This approach is to train a mapping from class A images to class B based on GANs. It is often used for entertainment tasks such as image style transfer. For example, Zhao et al. propose a GAN-based method with adversarial consistency loss for selfie to anime and other similar tasks [20]. If the image is classified as normal or abnormal, and a mapping function from abnormal image to normal one is trained, the abnormal region can be obtained by comparing the images before and after mapping, and the pixel-level segmentation can be realized. Siddiquee et al. devises a Fixed-Point network that utilizes weak label information, resulting in a more robust human disease detection capability than f-AnoGAN since the image is pre-categorized as normal or abnormal [21]. Ren et al. create a PAD-Net for detecting pavement distress, which enhances the detection ability of fine objects and achieves pixel-level distress segmentation [16].

Despite these advancements, fully unsupervised pixel-level solutions for pavement distress segmentation remain limited. The discussed weakly supervised methods utilize image-level annotations to guide network convergence, achieving pixel-level segmentation but still requiring a degree of supervision. This limitation emphasizes the need for further research into unsupervised methods that can effectively address the challenges specific to pavement distress detection.

C. Self-Supervised Learning Without Manual Annotations and Knowledge Gaps

Self-supervised learning offers an effective tool to achieve unsupervised representations by designing pretext tasks that generate artificial labels, enabling feature learning under unsupervised conditions. Wu et al. develop an instance discrimination task, where an image and its argumentations are considered positive samples while all other images are negative ones, delineating the direction for network optimization [33]. Tian et al. propose a contrast learning method by maximizing mutual information between different views of the same scene [34]. Pathak et al. train context encoders by contrasting an image and the same image with missing regions [35]. More recently, a unified scheme that predicts different views of the same image via Siamese networks has emerged [23], [25]. However, these generic pretext tasks may not be optimal for "long and strip" pavement images with scattered foreground objects, and mere deep representation is insufficient for pixel-level segmentation. Consequently, three critical questions arise in developing self-supervised learning for pavement distress segmentation: (i) how to automatically generate pseudo-labels from pavement images? (ii) how to train a deep neural network using these potentially noisy pseudo-labels? and (iii) how to utilize the trained deep neural network for segmentation?

Addressing these challenges requires leveraging domain-specific characteristics, such as the spatial contexts between continuous pavement images. The design of task-specific pretext tasks is crucial, as they should be customized to each computer vision application to exploit inherent data structures effectively.

III. Spatial Contexts-Informed Self-Supervised Pavement Distress Segmentation

The proposed scheme builds upon self-supervised learning but addresses the limitations of existing approaches, including: (i) proposing a spatial contexts-informed pretext task leverages consistent semantics across continuous pavement images; (ii) introducing open-ended multi-line parallel networks to exploit various noisy pseudo labels; and (iii) achieving distress segmentation in a fully unsupervised condition through the incorporation of the self-attention mechanism.

A. Spatial Contexts-Informed Pretext Task

Consider the set of continuous image frames captured by the pavement detection vehicle as $\{x_t\}$, where x_t represents the image captured at time t. When the time interval between x_{t_1} and x_{t_2} is sufficiently small, it can be assumed that they contain similar semantic information. This characteristic can be termed as spatial contexts. These similar images are then input to the network N for deep representations, which can be express as $N(x_{t_1}) = z_1, N(x_{t_2}) = z_2$. Valuable deep representations can be learned by optimizing the network N by minimizing loss function $L(z_1, z_2)$. Specifically, a pavement image x_t obtained at any given time can be expressed as $x_t = (\alpha_1, \alpha_2, \alpha_3, \cdots \beta_1, \cdots)$, where α_i represents the normal areas and β_i represents the abnormal distress areas on the pavement. Given that the pavement is constructed to uniform standards, the normal pavement should be consistent. Therefore, the network N should focus on the abnormal areas. Highlighting the focused areas can help to segment the abnormal areas, which in this case are pavement distress.

Contemporary datasets of pavement images in practical applications typically consist of entire road that have been segmented by mileage. The dataset used in this paper comprises 1,000 images sorted by mileage, each with a size of 3008 × 2048. As shown in Figure 1, a sliding window is designed with a size of 256 × 256 and a stride of 64 to

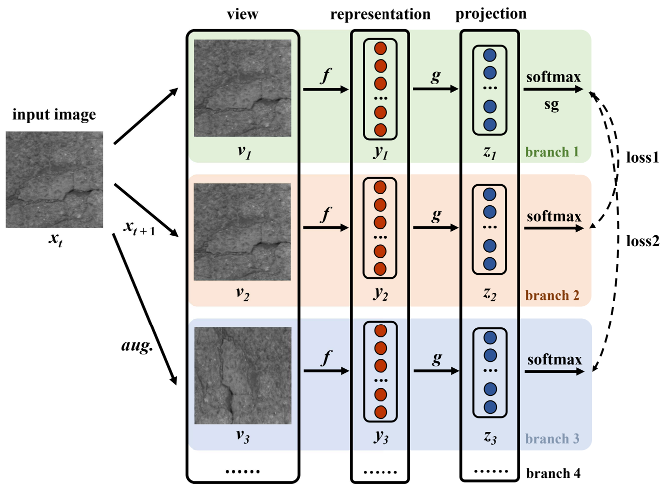


Fig. 2. Training pipeline of multi-line parallel network.

simulate the image frames. Specifically, it will intercept the image frames x_t from the original dataset images X_i in turn and name them according to the two-dimensional coordinates. Considering the calculation cost, a queue is employed to store the large images X_i and X_{i+1}. They are contacted into image X_c during obtaining the frames. X_i exits the queue when the sliding window is completely across it, then X_{i+2} joins the queue, and so on. It can avoid the dismiss of information at the connect areas of each image. After this preprocessing, about 1 300 000 image frames are obtained. Adjacent frames can be located according to their coordinates. For example, an image x_t with coordinate (m, n) has 4 adjacent images, which are image $x_{(m-1,n)}$, $x_{(m+1,n)}$, $x_{(m,n-1)}$, and $x_{(m,n+1)}$. Considering their similar semantic information, the spatial contexts-informed pretext task is proposed. The aim is to learn a mapping function $N(x_t) = z$ to implement a deep representation of the feature space and evaluate the spatial contexts from the parallel models as $d = S(N(v_1), N(v_2))$, where S is a metric function. v_1 and v_2 are the different views of the input image x_t. Here the direction of convergence of the deep leaning model is artificially created by minimizing $d = S(N(v_1), F(v_2))$, it enables an unsupervised training pipeline without any manual labeling.

B. Multi-Line Parallel Network for Various Pretext Tasks

To incorporate the proposed spatial contexts-informed pretext task into deep representation learning, we extend the Siamese Network into an open-ended multi-line parallel network. This architecture leverages spatial contexts-informed pseudo-labels while remaining flexible for integrating additional pretext tasks, as shown in Figure 2. This structure enables the input data to acquire deep representations within the same batch and complete the evaluation after passing through the projection layer. Taking the case where the batch size is 1 as an example, the deep representation $y = f(v)$ is initially obtained through the backbone network, utilizing different views v. Subsequently, $z = g(y)$ is derived through a multi-layer perceptron (MLP). These projection vectors undergo softmax processing, and the loss is calculated independently at the main view x_t. As shown in Figure 2, the latent representation obtained from the view that leaves the input image unchanged after forwarding the network is considered the label, while the latent representations obtained from the remaining views after forwarding the network serve as predictions. Subsequently, evaluating the difference between the label and the prediction through cross-entropy loss is a feasible solution:

$$\mathcal{S}(z_1, z_2) = -z_2 \log z_1 \tag{1}$$

Indeed, this constitutes a mutual prediction problem where any view can be treated as a label. Consequently, the calculation can be expressed as follows:

$$\mathcal{L}(z_1, z_2) = \frac{1}{2} S(z_1, z_2) + \frac{1}{2} S(z_2, z_1) \tag{2}$$

The final loss function is obtained by accumulating the different views and the original input image:

$$L_{total} = L_1(v_1, v_2) + L_2(v_1, v_3) + \cdots + L_i(v_1, v_{i+1}) \tag{3}$$

The standard ViT [32] is chosen as the backbone network $f(v)$, with its classifier removed. All parallel networks employ an identical architecture based on the same backbone network, initialized randomly. In branch 1 of the main view, parameters are not updated through backpropagation. Instead, a stop gradient strategy is employed, and parameters are updated through a momentum encoder from other branches that utilize normal backpropagation. This process is encapsulated by the following equation:

$$\theta_t = \alpha\theta_{t-1} + (1 - \alpha)\theta_t \tag{4}$$

where θ is the network parameter, t represents the iteration step, and α is the smoothing parameter that controls the moving average of the momentum encoder.

C. Unsupervised Segmentation via Self Attention

The backbone network trained by parallel networks has acquired the ability to represent the image frame x_t and its different views, which contain similar semantic information. It directs attention either towards regions of normal pavement, serving as the background, or towards regions of pavement distress, signifying the foreground. Pixel-level segmentation of pavement distress can be realized by revealing areas of focus or neglect by the backbone network. Following self-distillation with no labels (DINO) [36], when utilizing ViT as the backbone network, the input image is initially encoded into vectors through linear projections:

$$z_i = x_p^i E, \quad E \in \mathbb{R}^{(P^2 \bullet C) \times D} \tag{5}$$

These vectors, together with position embeddings and classification token, make up the input to the ViT encoder:

$$y_0 = [x_{class}; z_i;] + E_{pos}, \quad E \in \mathbb{R}^{(N+1) \times D} \tag{6}$$

y_0 will be normalized and put into the MSA block and MLP block:

$$y_1' = \mathrm{MSA}\,(\mathrm{LN}(y_0)) + y_0 \tag{7}$$

$$y_1 = \mathrm{MLP}\left(\mathrm{LN}(y_1')\right) + y_1' \tag{8}$$

Here MSA performs the following calculations:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

In the above equation, softmax$(QK^T / \sqrt{d_k})$ denotes the attention scores. For the one-dimensional classification token, its attention scores indicate the importance of each patch in measuring the semantic information about the consistency of the input image, considering the spatial contexts-informed pretext task. By visualizing the attention scores, a self-attention map can be created. Semantic segmentation images can then be obtained through simple thresholding.

D. Other Details

Following some settings from related contrastive learning work [37], [38], the centering is set up for each branch updated via backpropagation. Two A40 GPU is employed for the experiments, with a batch size of 64. AdamW is utilized as the optimizer and the learning rate is set to 0.0005. The momentum parameter is set to 0.996.

IV. Experiments

We conducted a comprehensive series of experiments to evaluate the effectiveness of the proposed spatial context-informed pretext task and the open-ended multi-line parallel network architecture in pavement distress segmentation. The proposed method requires the utilization of continuous pavement image characteristics. Although images captured under standard conditions are generally sequential, only a few available datasets meet this requirement. Consequently, our experiments were conducted using the previously mentioned continuous pavement dataset, Copave, which satisfies these criteria. Our experimental analysis centers on three key aspects:

(i) The comparative segmentation accuracy of our spatial context-informed method against other pretext tasks.

(ii) The potential performance gains of the proposed open-ended multi-line parallel network when integrated with multiple pretext tasks.

(iii) The segmentation accuracy of our fully unsupervised methods relative to weakly supervised approaches in the context of pavement distress segmentation.

A. Experiment 1: Evaluating Spatial Context-Informed Pretext Task Against Existing Methods

To quantitatively assess the segmentation accuracy of our proposed spatial context-informed pretext task in comparison with state-of-the-art self-supervised learning methods, we evaluate its task-specific suitability and efficacy for pavement distress segmentation.

1) Baselines: Three state-of-the-art approaches are chosen as baseline models, including:

(i) Masked autoencoders (MAE) [26]. As depicted in Figure 3(a), MAE employs an masked image reconstruction pretext task to train the backbone and obtain a deep representation of the image.

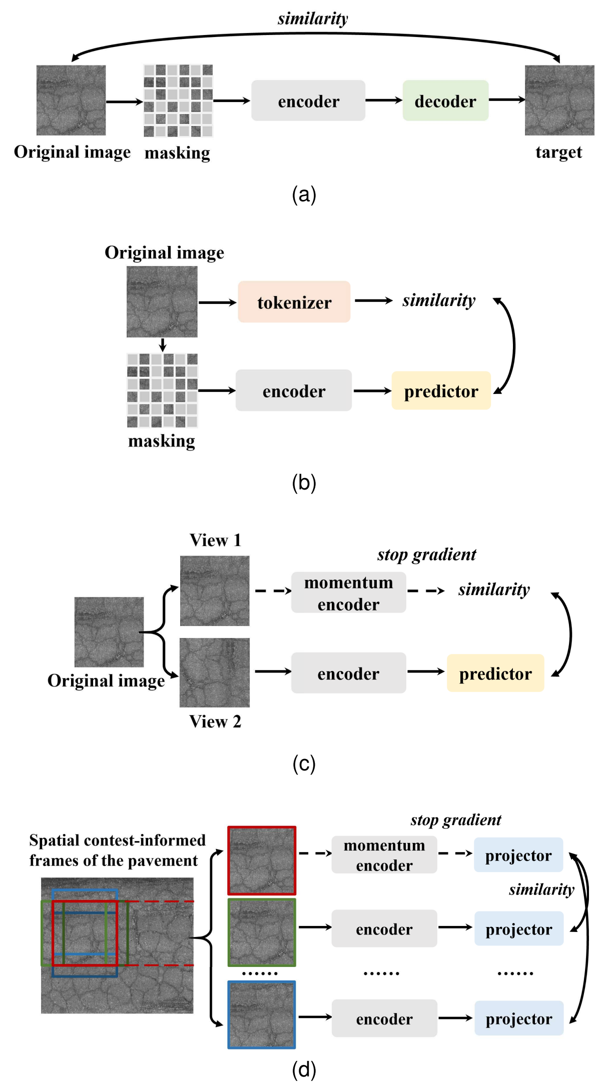


Fig. 3. Training pipeline of different approaches: (a) MAE; (b) BEIT; (c) MoCo v3; and (d) Ours.

(ii) Bidirectional encoder representation from image Transformers (BEIT) [39]. As depicted in Figure 3(b), the backbone is trained by comparing the visual token obtained from the masked image block through the BEIT encoder with the visual token obtained from the trained image tokenizer.

(iii) Momentum contrast for unsupervised visual representation learning version 3 (MoCo v3) [40]. As depicted in Figure 3(c), MoCo v3 utilizes parallel networks to train the backbone by comparing the deep representations of various views of the same input image in each network.

2) Experiment Settings: This experiment implements unsupervised visual representation learning through self-supervised learning. ViT-small networks serve as the backbone for all models, and the training is conducted on the pavement image dataset created in this paper, consisting of 1,300,000 image frames. The hyperparameter settings for the three baseline models adhere to the specifications outlined in the original paper. The self-attention maps are obtained by visualizing the attention scores of the classification token using the trained ViT backbone networks. The proposed method deviates from

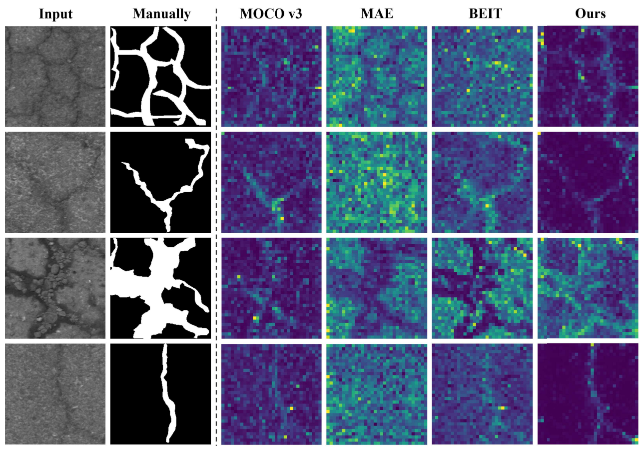


Fig. 4. Self-attention maps for different pretext tasks.

TABLE I
EVALUATION METRICS FOR DIFFERENT PRETEXT TASKS

Method	mIoU	mDice	mPA
MAE	0.467	0.602	0.683
MoCo v3	0.440	0.575	0.688
BEIT	0.414	0.556	0.630
Ours	**0.515**	**0.635**	**0.778**

the baseline solely in terms of pretext task training. Each method undergoes training for 20 epochs.

3) Results and Analysis: Figure 4 presents the self-supervised attention maps generated by the three baseline models (MoCo v3, MAE, and BEIT) and the proposed method, all trained from scratch using random weight initialization. For each approach, the self-attention head yielding the best performance is selected for visualization. The outcomes reveal that all methods exhibit varying capabilities in attending to pavement distress regions. Compared to the baseline models, the proposed method more effectively discriminates between distressed foreground areas and the background pavement, suggesting that the classification token employed in the proposed approach is more adept at identifying distress regions. MoCo v3 primarily focuses on a few anomalous areas (yellow and green), which are not clearly distinguishable from the background pavement. MAE tends to assign higher attention scores to background pavement regions while underemphasizing obvious distress areas. BEIT exhibits an alternating behavior, attending to distress regions in some instances (second row of Figure 4) and background areas in others (third row of Figure 4). Additionally, it demonstrates a tendency to emphasize extensive distress patterns while potentially overlooking finer crack regions.

After obtaining the attention scores for each image patch, semantic segmentation results are derived through simple thresholding at a value of 0.5 across all methods. Manually labeled images serve as the ground truth for evaluation. Three metrics, namely mean Dice coefficient (mDice), mean Intersection over Union (mIoU), and mean Pixel Accuracy (mPA), are selected to assess segmentation performance. As illustrated in Table I, the results reveal that the proposed method attains optimal performance across all evaluation indicators. Taking the mIoU metric as an example, the proposed approach at least improves the segmentation accuracy by an absolute value of 0.048 compared to the baseline models.

The reasons for the poor performance of the three baseline methods are different, which is highly related to their pretext task. For example, the mask setting in MAE might cover the pavement distress areas, particularly when the distresses are fine cracks or potholes. Reconstructing the original image from the remaining portion may prove challenging due to the lack of meaningful information in most of the pavement background pixels. The ability of MAE to reconstruct the original image even with a high percentage of mask ratio (e.g., 75%) is a major contribution in the original paper, and it yielded surprisingly good results. However, in our task, we tested mask ratios of 25%, 50%, and 75%, with the best outcomes achieved at a 50% mask ratio. Therefore, the findings in Table I do not align with the initial 75% mask ratio configuration. Figure 4 provides additional evidence, showcasing that MAE prioritizes the background area of the image over the distress area.

MoCo v3 and the proposed method share similarities in terms of model construction. However, the pretext task in MoCo v3 involves predicting different views of the same image using conventional image augmentation techniques such as random cropping, flipping, and color jittering. These techniques do not fully exploit the benefits of continuous shooting of pavement images. In contrast, the proposed method can apply the MoCo v3 pretext task while also incorporating the spatial context-informed pretext task due to their similarity in model construction. A detailed analysis will be presented in section IV-B.

The poor performance of BEIT primarily stems from its unstable attention allocation. It utilizes a self-supervised training model similar to BERT, a natural language processing method, and introduces a pretext task called masked image modeling (MIM). The main difference between BEIT and MAE is the use of the image tokenizer, enabling training based on comparing visual tokens. The training process for BEIT is more time-consuming than MAE. Since this encoder incorporates out-of-domain data rather than solely relying on pavement images, the model may concentrate its attention on foreground regions and, in other cases, focuses on background regions due to the nature of the MIM pretext task. This inconsistency is particularly evident when detecting small cracks. Figures 4 further illustrate that while BEIT performs relatively well on large-scale defects, its segmentation accuracy is markedly insufficient for fine-grained defects such as alligator cracks.

B. Experiment 2: Multi-Task Integration in Open-Ended Multi-Line Parallel Networks

To systematically evaluate the performance of our open-ended multi-line parallel network architecture in integrating multiple pretext tasks, we quantify its ability to leverage diverse self-supervised signals and examines their impact on segmentation accuracy.

1) Baselines: The proposed flexible multi-line parallel network is designed to integrate multiple semantically related pretext tasks. As illustrated in Figure 5, the experiments in this section investigate the combination of the spatial

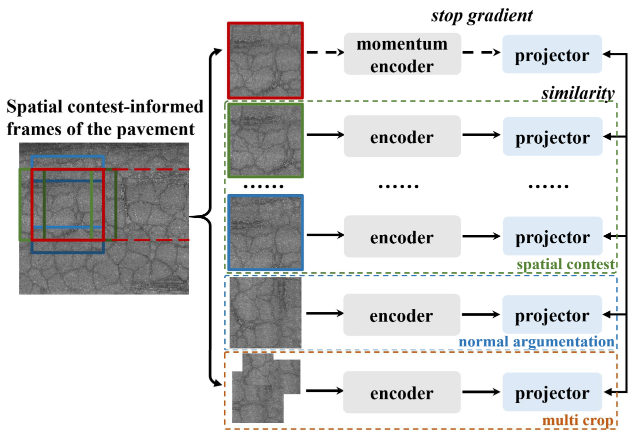


Fig. 5. Training pipeline for combining different pretext tasks.

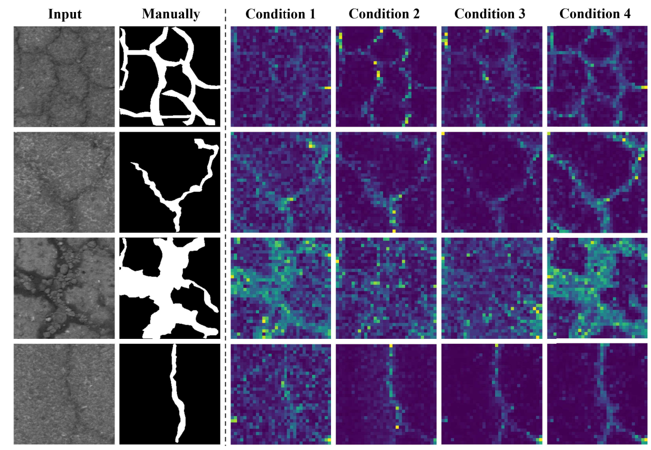


Fig. 6. Self-attention maps for various pretext task combinations.

TABLE II

EVALUATION METRICS FOR VARIOUS PRETEXT TASK COMBINATIONS

Con.	SC	RDA	Multi-crop	mIoU	mDice	mPA
1		✓		0.440	0.575	0.688
2	✓			0.515	0.635	0.778
3		✓	✓	0.490	0.629	0.721
4	✓	✓	✓	**0.583**	**0.706**	**0.827**

context-informed (SC) task with regular data augmentation (RDA) techniques, such as random flipping, color jitter, and random grayscale [23], as well as the multi-crop strategy [24], which involves representing the entire image using smaller, localized patches. Multi-crop is commonly applied alongside traditional augmentation methods and has demonstrated effectiveness in general visual representation learning. All three tasks are suitable for self-supervised learning, as they preserve semantic consistency with the original input and can be compared against it during training. Following this process, the trained network follows the testing method outlined in Section III-C to obtain unsupervised segmentation results.

2) Experiment Settings: This section utilizes the same encoder (ViT small) and loss function (described in Section III-B) across all methods for consistency. The same dataset containing 1.3 million images was used for training, with each method trained for 20 epochs. Pixel-level segmentation is performed using the approach outlined in Section III-C. The key difference lies in the self-supervised pretext tasks employed during training, which include four combinations: (1) using only regular data augmentation methods; (2) using only the proposed spatial context-informed task; (3) combining regular data augmentation methods with multi-crop augmentation; and (4) using the spatial context-informed task along with regular data augmentation methods and multi-crop augmentation.

3) Results and Analysis: Table II presents the quantitative results under four experimental conditions. The outcomes from Conditions 1 and 2 illustrate the adaptability of both the regular data augmentation (RDA) method and the proposed spatial context-informed (SC) task for pavement distress segmentation. The SC task outperformed RDA, achieving 0.515 mIoU, 0.635 mDice, and 0.778 mPA, representing improvements of 17.0%, 10.4%, and 13.1%. When the multi-crop strategy was introduced to the RDA method in Conditions 3, it exhibits notable performance gains. However, a substantial performance gap remains when the SC task was excluded. As shown in Condition 4, the proposed method achieved 0.583 mIoU, 0.706 mDice, and 0.827 mPA, representing improvements of 19.0%, 12.2%, and 14.7%. The comparison between Condition 2 and Condition 4 underscore the benefit of the flexible multi-line parallel network architecture, which enables the integration of the SC task with other compatible pretext tasks. As a result, the segmentation performance was further enhanced by 0.068 mIoU, 0.071 mDice, and 0.049 mPA, representing improvements of 13.2%, 11.2%, and 6.3%. These results highlight two key strengths of the proposed method. First, the spatial context-informed task is well-suited for pavement distress segmentation. Second, the scalable multi-line parallel architecture supports the seamless integration of related self-supervised tasks, leading to consistently improved segmentation performance.

Figure 6 illustrates the self-attention maps for various pretext task combinations. A comparison between condition 1 and condition 2 reveals that the spatial context-informed task exhibits superior discrimination between foreground and background. This distinction is evident in the lower self-attention score for the background, depicted in dark blue. Similarly, when contrasting condition 2 with condition 3, it becomes apparent that the spatial context-informed task outperforms the combined pretext task involving normal data augmentation and multi-crop. This is evident in the higher self-attention score for the foreground patch, represented by a bright yellow/green color. Condition 4 showcases our recommended optimal solution, amalgamating normal data augmentation, multi-crop, and the spatial context-informed task. This solution effectively distinguishes between foreground and background areas with clarity and precision.

C. Experiment 3: Benchmarking Against Weakly Supervised Frameworks

To benchmark our unsupervised approach against leading weakly supervised methodologies for pavement distress segmentation, we empirically evaluate the viability of our label-free method as a practical alternative to approaches that rely on image-level annotations.

1) Baselines: Given the scarcity of weakly supervised or unsupervised methods explicitly tailored for pavement distress segmentation, this section conducts a comparison with state-of-the-art anomaly detection methods commonly applied in related domains, including human disease detection and industrial anomaly detection. These methods are categorized into three groups based on the training pipeline, as illustrated in Figure 7.

(i) Weakly supervised learning based on image reconstruction: The first step involves selecting normal images from the original pavement dataset through image-level labeling. Subsequently, a reconstruction network is trained with the assumption that even for abnormal input images, the network can generate a normal output. The identification of the distress region is based on comparing the discrepancies between the input and output images. Three commonly used methods are employed in this experiment, including GANomaly [17], f-Anogan [18], and FastFlow [32].

(ii) Weakly supervised learning based on image-to-image translation: Initially, the original pavement dataset is categorized into two groups, normal and abnormal, guided by image-level labels. Subsequently, a mapping function is trained to transform abnormal images into a normal state. The detection of the distress area involves comparing the differences observed before and after the transformation. Three commonly used methods are employed in this experiment, including ACl GAN [20], Fixed Point [21], and PAD Net [16]. Notably, PAD Net is specifically designed for weakly supervised pavement distress detection.

(iii) Self-supervised learning: The proposed method and selected baseline approaches, including MAE [26], BEIT [39] and MoCo v3 [40] (as mentioned in Section IV-A), involve identifying distress regions through the utilization of self-attention mechanisms. This is facilitated by formulating pretext tasks and executing self-supervised training procedures.

2) Experiment Settings: Since this section involves completely different methods compared to the previous experiments, the recommended parameters for each weakly supervised method are followed, and training is performed until convergence. All methods utilize the same dataset, which is identical to the one used in Experiment 1 and Experiment 2. As weakly supervised methods require image-level labels, 400 images were manually labeled as either normal or abnormal, and a ResNet-101 classifier was trained to obtain these image-level labels for the dataset.

3) Results and Analysis: Figure 8 illustrates the qualitative outcomes of the experiments. FastFlow [32], f-Anogan [18], and GANomaly [17] are classified as class A methods, while ACl GAN [20], Fixed Point [21], and PAD Net [16] are classified as class B methods. MAE [26], BEIT [39], MoCo v3 [40] and the proposed approach are classified as class C methods. The approaches tailored specifically for pavement distress segmentation, such as PADNet and the proposed method, demonstrated a distinct advantage by being capable of approximately delineating the contours of cracks and potholes.

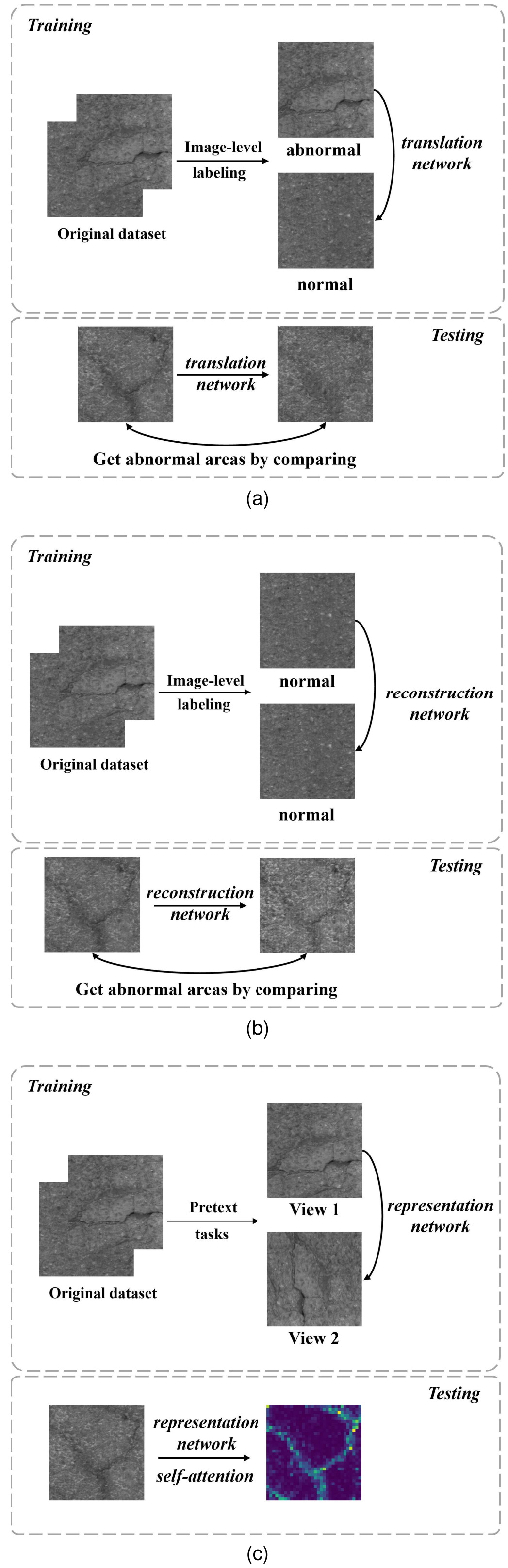


Fig. 7. Three types of training and testing for weakly supervised or unsupervised pavement distress segmentation methods. (a) weakly supervised learning based on image reconstruction; (b) weakly supervised learning based on image-to-image translation; and (c) self-supervised learning.

This observation may stem from the low contrast of pavement images captured in the field, as opposed to industrial or medical images that typically exhibit a more standardized

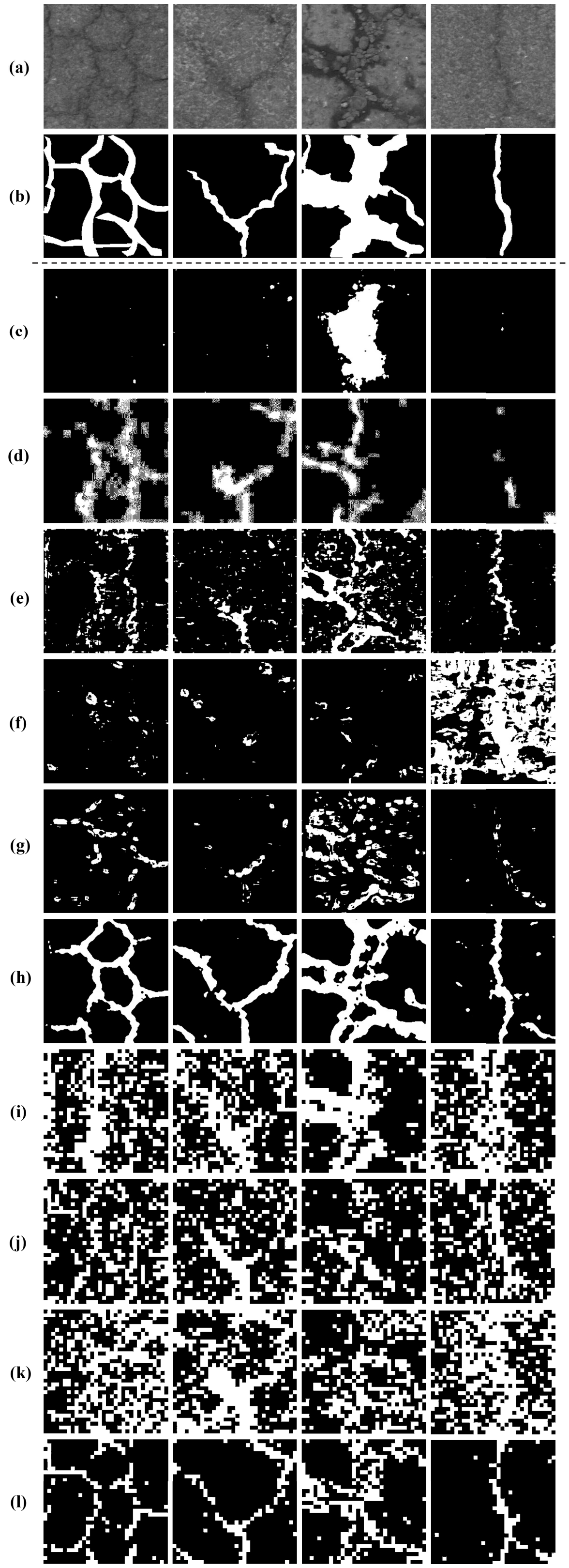


Fig. 8. Segmentation results on baseline and proposed methods. (a) Input image. (b)Manual label. (c) FastFlow. (d) GANormaly. (e) f-Anogan. (f) ACL GAN. (g) Fixed Point. (h) PAD Net. (i) MAE. (j) MoCo v3 (k) BEIT (l) Ours.

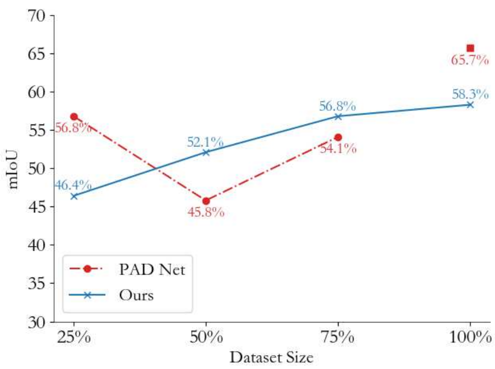
and consistent shooting environment. Despite the comparison being against weakly supervised schemes, the results achieved by the proposed method are competitive.

TABLE III
EVALUATION METRICS FOR VARIOUS PRETEXT TASK COMBINATIONS

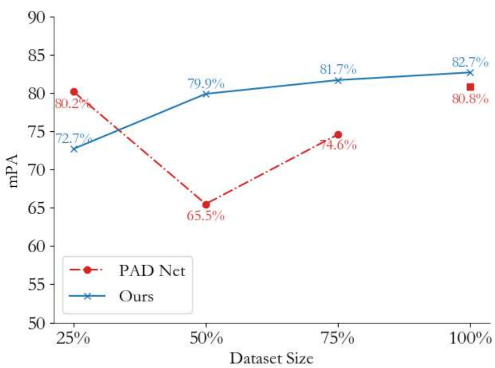
Methods	**Categories**	**mIoU**	**mDice**	**mPA**
FastFlow	A	0.464	0.536	0.784
GANormaly	A	0.538	0.655	0.787
f-Anogan	A	0.442	0.525	0.784
ACL GAN	B	0.450	0.542	0.770
Fixed Point	B	0.497	0.599	0.707
PAD Net	B	**0.657**	**0.772**	0.808
MAE	C	0.467	0.602	0.683
MoCo v3	C	0.440	0.575	0.688
BEIT	C	0.414	0.556	0.630
Proposed	C	**0.583**	**0.706**	**0.827**

The quantitative results of the experiments are presented in Table III. Among the evaluated schemes, the proposed method demonstrates superior performance except for PAD Net in mIoU and mDice metric, but the proposed approach exhibits a slight advantage in the mPA metric. Remarkably, the proposed method surpasses the state-of-the-art unsupervised learning methods for general vision tasks, achieving a substantial improvement of 0.143 when evaluated using the mIoU index, as measured by the average accuracy across the three methods. It is evident that different categories of methods exhibit varying sensitivities to various evaluation metrics. The direct application of weakly supervised methods, including Class A and Class B, presents a trade-off: while these methods are disadvantaged when prioritizing evaluation metrics such as the intersection over union ratio (IoU), they exhibit an advantage when favoring evaluation metrics like pixel accuracy (PA). This phenomenon can be attributed to the inherent tendency of weakly supervised methods to regard objects that are challenging to detect as background. Consequently, pavement distresses often occupy a relatively small proportion of the image, these methods can achieve high mPA scores even when failing to detect all distresses. In contrast, self-supervised methods (Class C) are predisposed to identifying the foreground region, rendering them more susceptible to misclassifying the background region.

While the performance difference between the proposed method and the state-of-the-art weakly supervised schemes like PAD Net is marginal, the proposed approach demonstrates its own merits. Figure 9 illustrates the outcomes obtained when training PAD Net and the proposed method on datasets of varying sizes. When trained on the entire dataset (100%), PAD Net encountered a gradient vanishing issue, resulting in premature termination. This problem can be attributed to the imbalance between positive and negative samples, with 1,066,130 normal images and only 259,473 abnormal images. The red square points in Figure 9 represent the results obtained by training on a dataset generated by cropping 1000 high-resolution images into smaller ones, rather than using sliding windows (followed as [16]). This cropping strategy offers the advantage of creating a relatively balanced distribution of positive and negative samples, yielding favorable outcomes. A comparative analysis of these two competitive methodologies reveals the following observations: First, the proposed method exhibits a



(a)



(b)

Fig. 9. Results of training PAD net and the proposed method using datasets of different sizes, with different metrics of (a) mIoU, (b) mPA.

progressive increase in performance as the dataset size grows, ultimately achieving optimal results constrained by the dataset scale. In contrast, PAD Net exhibits an erratic convergence behavior and consistently underperforms in terms of the mPA metric. Second, as the dataset size expands, fully unsupervised methods can operate unhindered, while weakly supervised methods require additional manual annotations, potentially leading to sample imbalances and subsequently impacting detection outcomes.

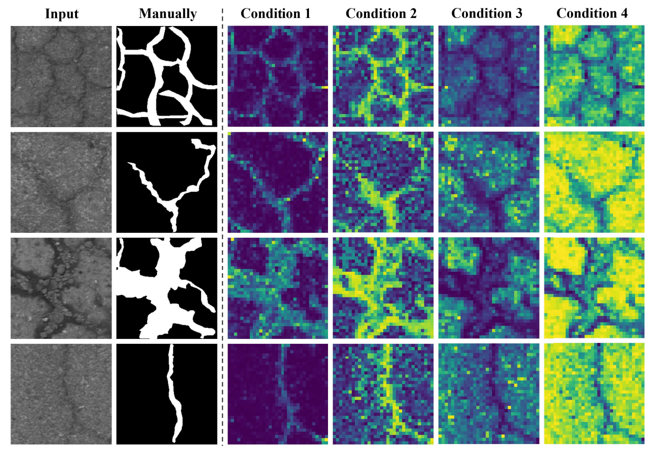


Fig. 10. Self-Attention maps for different settings.

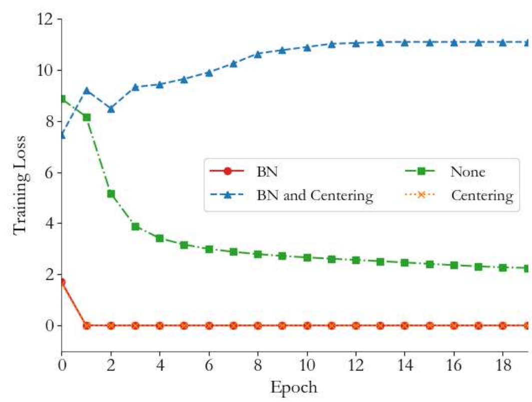


Fig. 11. Comparison of loss convergence in different settings.

TABLE IV
EVALUATION METRICS FOR VARIOUS PRETEXT TASK COMBINATIONS

Con.	BN	Centering	mIoU	mDice	mPA
1		✓	**0.583**	**0.706**	**0.827**
2	✓	✓	0.551	0.676	0.776
3	✓		0.550	0.677	0.769
4			0.548	0.676	0.768

V. DISCUSSION

While deep representations based on self-supervised learning have achieved remarkable success in general computer vision tasks, their application to pixel-level segmentation remains an emerging area, especially for targets like pavement distress dominated by small, elongated, and irregularly shaped objects such as cracks. This has necessitated the exploration of specialized methods for pretext tasks of creating pseudo labels. The proposed spatial context-informed pretext task can be viewed as analyzing a video sequence of continuously captured pavement images. As the frames advance, the human visual system can perceive a motion effect akin to moving pavement distress, allowing contextual information to be exploited. Experimental results demonstrate that the proposed scheme is better suited for pavement distress detection compared to mask-based approaches like MAE and BEIT that have achieved success in general vision tasks. Consequently, the proposed multi-line parallel network incorporates data augmentation techniques that preserve complete image information, rather than including mask-based schemes. Moreover, compared to weakly supervised models, the proposed unsupervised approach offers key advantages beyond competitive accuracy: (i) it does not require balanced label distributions, and its performance can continually improve with increasing data volumes; (ii) the approach is highly scalable, allowing the integration of diverse pretext tasks; and (iii) the underlying transformer architecture provides opportunities to enhance performance by effectively fusing multimodal information sources beyond just visual data.

However, it does not imply that self-supervised learning networks can converge effectively without additional measures. To enable the network to accurately detect distress targets, some necessary configurations are adopted. This paper follows a variety of different practices in related works, such as batch normalization (BN) and centering, that were correlated with the convergence of the proposed method. Four combinations

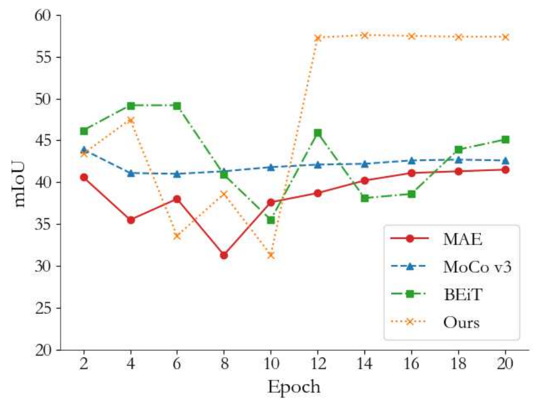


Fig. 12. Evaluation of different pretext task during trainin.

of these techniques are experimented, as shown in Table IV. Figure 10 displays the self-attention map, and Figure 11 shows the corresponding convergence of the loss. The results in Table IV indicate only minor differences but the self-attention maps reveal significant distinctions. It is observed that without centering, the self-attention layers concentrate more on the background pavement region rather than the foreground distress region. Consequently, when centering is not used, the opposite threshold setting, i.e., removing the more focused regions are employed. When both BN and centering are used, the self-attention layer becomes more sensitive to changes in the pavement, shown as the distress regions receiving higher attention scores, while some of the pavement regions are equally attended to. This makes the self-attention map noisy and unsuitable for unsupervised pavement distress segmentation. A more notable difference appears in the convergence of the loss in Figure 11, which fails to converge well when both BN and centering are used. Considering the above, only centering as the setting in this paper is employed. It obtains the best mDice and mPA and exhibits a better self-attention map and loss convergence process. Unfortunately, the theoretical underpinnings for these various configurations remain a subject of debate, their effectiveness can vary across different tasks.

Beyond this, a natural consideration is whether pre-training the model on a large common vision dataset could enhance its performance. To investigate this, the methods in section IV-A were evaluated using weights from a ViT-small backbone network pre-trained on the ImageNet dataset followed by fine-tuning on the pavement dataset using the described methodologies. Figure 12 illustrates the progression of the mIoU evaluation metric for different methods across training epochs. Counterintuitively, as presented in Table I, all four methods underperformed compared to training from scratch. It was observed that the performance of these methods did not improve with continued training, suggesting that this phenomenon is not unique to the proposed approaches but rather inherent in self-supervised learning methods. A potential explanation for this outcome may lie in the substantial domain discrepancy between the ImageNet dataset, consisting of natural images, and the pavement dataset, consisting of specialized infrastructure images. Moreover, the backbone network obtained through supervised training on a classification task may differ fundamentally from those acquired through unsupervised learning objectives. These findings underscore the necessity of a large, domain-specific dataset and a tailored training strategy to effectively address pavement distress detection tasks.

Although the proposed method has achieved competitive results on the task of unsupervised pavement distress segmentation, there are still some limitations. Due to dataset and computational resource constraints, this study, as an initial exploration, failed to fully capture the deepest potential of the proposed method. Based on the experience with existing large multimodal models, the proposed approach may achieve more competitive results as data size and model capacity increase. On the other hand, under limited equipment and data scenarios, efficient, rapid, and lightweight model training is also a problem worth exploring.

VI. Conclusion

Automated and intelligent distress detection is a crucial enabler for future transportation infrastructure maintenance. To circumvent the time-consuming and labor-intensive manual annotation required by deep learning-based pavement distress segmentation methods, this paper explores a novel self-supervised learning approach that eliminates the need for manual labeling. The authors leverage the spatial context present in continuous pavement images to create pseudo-labels and design an open-ended multi-line parallel network to learn deep representations using these labels, ultimately segmenting distress regions via self-attention mechanisms. Experimental results demonstrate that: (1) the proposed spatial context-informed pretext task achieves an average 0.075 mIoU higher accuracy than state-of-the-art approaches such as MAE, BEIT, and MoCo v3 on the pavement distress segmentation task; (2) the proposed open-ended multi-line parallel network exhibits the potential to integrate multiple pretext tasks, with the combined scheme successfully improving detection accuracy by 0.068 mIoU; (3) the proposed unsupervised method remains competitive with weakly supervised approaches requiring image-level labels, surpassing most weakly supervised techniques. The authors acknowledge that the performance of the proposed approach is currently constrained by model and dataset size limitations but believe that these findings will contribute to the development of fully automated pavement distress detection systems, with substantial performance enhancements achievable by scaling these factors.

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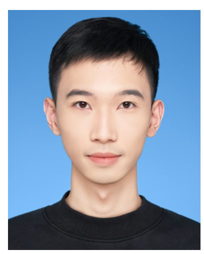

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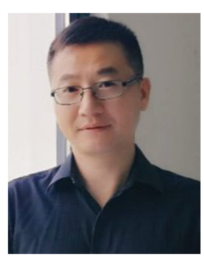

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